

Consensus molecular subtyping of metastatic colorectal cancer expands biomarker-directed therapeutic benefit for patients with CMS1 and CMS2 tumors.

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We developed a whole transcriptome sequencing (WTS)-based Consensus Molecular Subtypes (CMS) classifier using FFPE tissue and investigated its prognostic and predictive utility in a large clinico-genomic database of CRC patients ($n = 24,939$). The classifier was trained against the original CMS datasets using an SVM model and validated in an independent blinded TCGA dataset (88.0% accuracy). Kaplan-Meier estimates of overall survival (OS) and time-on-treatment (TOT) were calculated for each CMS ($p < 0.05$ considered significant). CMS2 tumors were enriched on left-side of colon and conferred the longest median OS. In RAS-wildtype mCRC, left-sided tumors and CMS2 classification were associated with longer TOT with anti-EGFR antibodies (cetuximab and panitumumab). When restricting to only CMS2, there was no significant difference in TOT between right- versus left-sided tumors. CMS1 tumors were associated with a longer median TOT with pembrolizumab relative to other CMS groups, even when analyzing only microsatellite stable (MSS) tumors. A WTS-based CMS classifier allowed investigation of a large multi-institutional clinico-genomic mCRC cohort, suggesting anti-EGFR therapy benefit for right-sided RAS-WT CMS2 tumors and immune checkpoint inhibitor benefit for MSS CMS1. Routine CMS classification of CRC provides important treatment associations that should be further investigated.

Consensus molecular subtypes classification of colorectal cancer as a predictive factor for chemotherapeutic efficacy against metastatic colorectal cancer.

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The consensus molecular subtypes (CMS) classification is one of the most robust colorectal cancer (CRC) classifications based on comprehensive gene expression profiles. This study aimed to clarify whether the CMS is a predictive factor for therapeutic effects of standard chemotherapies for metastatic CRC (mCRC). We retrospectively enrolled 193 patients with mCRCs, and using comprehensive gene expression data, classified them into 4 subtypes: CMS1-CMS4. The associations between the subtypes and treatment outcomes were analyzed. Regarding first-line chemotherapy, irinotecan (IRI)-based chemotherapy was significantly superior to oxaliplatin (OX)-based chemotherapy for progression-free survival (PFS; hazard ratio [HR] = 0.31, 95% confidence interval [CI] 0.13-0.64) and overall survival (OS; HR = 0.45, 95% CI 0.19-0.99) in CMS4. Regarding the anti-epidermal growth factor receptor (anti-EGFR) therapy, CMS1 showed particularly worse PFS (HR = 2.50, 95% CI 1.31-4.39) and OS (HR = 4.23, 95% CI 1.83-9.04), and CMS2 showed particularly good PFS (HR = 0.67, 95% CI 0.44-1.01) and OS (HR = 0.49, 95% CI 0.27-0.87) compared with the other subtypes. The biological characteristics of CMS may influence the efficacy of chemotherapy. CMS might be a new predictive factor for the efficacy of chemotherapy against mCRCs.

ColoType: a forty gene signature for consensus molecular subtyping of colorectal cancer tumors using whole-genome assay or targeted RNA-sequencing.

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Colorectal cancer (CRC) tumors can be partitioned into four biologically distinct consensus molecular subtypes (CMS1-4) using gene expression. Evidence is accumulating that tumors in different subtypes

are likely to respond differently to treatments. However, to date, there is no clinical diagnostic test for CMS subtyping. In this study, we used novel methodology in a multi-cohort training domain ($n = 1,214$) to develop the ColoType scores and classifier to predict CMS1-4 based on expression of 40 genes. In three validation cohorts ($n = 1,744$, in total) representing three distinct gene-expression measurement technologies, ColoType predicted gold-standard CMS subtypes with accuracies 0.90, 0.91, 0.88, respectively. To accommodate for potential intratumoral heterogeneity and tumors of mixed subtypes, ColoType was designed to report continuous scores measuring the prevalence of each of CMS1-4 in a tumor, in addition to specifying the most prevalent subtype. For analysis of clinical specimens, ColoType was also implemented with targeted RNA-sequencing (Illumina AmpliSeq). In a series of formalin-fixed, paraffin-embedded CRC samples ($n = 49$), ColoType by targeted RNA-sequencing agreed with subtypes predicted by two independent methods with accuracies 0.92, 0.82, respectively. With further validation, ColoType by targeted RNA-sequencing, may enable clinical application of CMS subtyping with widely-available and cost-effective technology.

Consensus molecular subtyping improves the clinical usefulness of canonical tumor markers for colorectal cancer.

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Transcriptome-based classification, such as consensus molecular subtyping, is expected to be applied to colorectal cancer (CRC). However, the relationship between molecular profiles and classical tumor markers, which are already used in clinical practice, has not been analyzed in a large cohort and remains unclear. We classified more than 1,500 Japanese patients with CRC based on consensus molecular subtyping and investigated the clinically available blood carcinoembryonic antigen (CEA) concentrations of each subgroup. To precisely distinguish CRCs, we allocated them to five subgroups, including tumors that were difficult to classify using the consensus molecular subtypes (CMSs), and extracted a heterogeneous population with somatic mutations and expression profiles that differed from those of the CMSs 1-4. For patients allocated to the CMS4 subgroup of stage III CRCs, elevated blood CEA concentrations may identify a subgroup with highly aggressive disease and contribute to improving therapeutic decisions. Furthermore, gene expression and pathway analyses of tumor and non-tumor tissues revealed that tumor immunity was "cold" in this subgroup with high CEA concentrations. The combination of emerging molecular profiling and classical tumor markers may have greater clinical utility than either used alone.

Colorectal cancer subtypes: Translation to routine clinical pathology.

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Colorectal cancer (CRC) is the second most common cause of cancer death in Europe. Although outcomes have improved, it is clear that from a genomic standpoint CRC is not one disease, but a heterogeneous group of malignancies that arise within one organ. Given that different subtypes have different outcomes, the ability to subtype tumours in the clinic would be highly favourable, enabling optimal treatment for individual patients. In 2015, a consortium proposed four consensus subtypes for CRC (MSI immune, canonical, metabolic, and mesenchymal) based on six classifications systems reported to have prognostic value. However, genomic assessment of tumours is not readily translated into routine pathology with a

need for standardisation and reproducibility of assessment. Immunohistochemistry is widely used in routine pathology, and would present a more readily translatable method for subtyping CRC tumours. Therefore, the literature was reviewed to characterise the genomic and phenotypic features associated with each subtype, with the aim of enabling subtyping of CRC to be taken forward into routine clinical practice.

A Novel Molecular Analysis Approach in Colorectal Cancer Suggests New Treatment Opportunities.

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Colorectal cancer (CRC) is a molecular and clinically heterogeneous disease. In 2015, the Colorectal Cancer Subtyping Consortium classified CRC into four consensus molecular subtypes (CMS), but these CMS have had little impact on clinical practice. The purpose of this study is to deepen the molecular characterization of CRC. A novel approach, based on probabilistic graphical models (PGM) and sparse k-means-consensus cluster layer analyses, was applied in order to functionally characterize CRC tumors. First, PGM was used to functionally characterize CRC, and then sparse k-means-consensus cluster was used to explore layers of biological information and establish classifications. To this aim, gene expression and clinical data of 805 CRC samples from three databases were analyzed. Three different layers based on biological features were identified: adhesion, immune, and molecular. The adhesion layer divided patients into high and low adhesion groups, with prognostic value. The immune layer divided patients into immune-high and immune-low groups, according to the expression of immune-related genes. The molecular layer established four molecular groups related to stem cells, metabolism, the Wnt signaling pathway, and extracellular functions. Immune-high patients, with higher expression of immune-related genes and genes involved in the viral mimicry response, may benefit from immunotherapy and viral mimicry-related therapies. Additionally, several possible therapeutic targets have been identified in each molecular group. Therefore, this improved CRC classification could be useful in searching for new therapeutic targets and specific therapeutic strategies in CRC disease.

Prognostic and predictive molecular biomarkers in advanced colorectal cancer.

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The revolution of precision medicine has produced unprecedented seismic shifts in the treatment paradigm of advanced cancers. Among the major killers, colorectal cancer (CRC) is far behind the others. In fact, the great successes obtained in breast, NSCLC, melanoma, and genitourinary tract tumors have been observed only in fewer than 5 % metastatic colorectal cancer (mCRC): those with the mismatch repair deficiency (dMMR), a well-known predictive factor for the outstanding efficacy of checkpoint inhibitors (CPI). The treatment of the remaining vast majority mCRC patients is still based upon only two molecular determinants: the RAS and BRAF mutational status. New promising biomarkers include HER2, tumor mutational burden (TMB) for its possible implications on CPI efficacy, and the extremely rare NTRK fusions. The Consensus Molecular Subtypes classification (CMS) is a good example of the efforts to combine different molecular features of this disease, although its relevance in clinical practice is still under investigation. In this Review, we focus on all these prognostic and predictive biomarkers, analyzing data from the most important clinical trials of the last years. We also try to rank them according to their

prognostic and predictive power.

Genetic Molecular Subtypes in Optimizing Personalized Therapy for Metastatic Colorectal Cancer.

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Colorectal cancer (CRC) is a heterogeneous disease entity in terms of both molecular carcinogenesis and morphologic carcinogenesis multistep pathways. Considerable heterogeneity exists within CRC due to the varied genetic and epigenetic mechanisms involved in different carcinogenesis pathways. A better understanding of pathophysiology of tumors is necessary to develop modern and successful means of treatment in metastatic CRC. Over the last 5 years, there has been a surge in interest in the molecular classification of colorectal cancer, as its clinical importance both for predicting prognosis and in guiding personalized treatment had been acknowledged. Recently, the Colorectal Cancer Subtyping Consortium identified four consensus molecular subtypes, CMS 1-4 in CRC; however, attempts to stratify CRC using molecular features for prognostic and predictive purposes in clinical conditions had limited success. In this review, we focused on molecularly defined subtypes of CRC including specific mutations and discuss implications for current and future patient management in metastatic CRC to achieve the maximal therapeutic response for each patient, while reducing adverse side effects of therapy.

Exploring and modelling colon cancer inter-tumour heterogeneity: opportunities and challenges.

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Colon cancer inter-tumour heterogeneity is installed on multiple levels, ranging from (epi)genetic driver events to signalling pathway rewiring reflected by differential gene expression patterns. Although the existence of heterogeneity in colon cancer has been recognised for a longer period of time, it is sparingly incorporated as a determining factor in current clinical practice. Here we describe how unsupervised gene expression-based classification efforts, amongst which the consensus molecular subtypes (CMS), can stratify patients in biological subgroups associated with distinct disease outcome and responses to therapy. We will discuss what is needed to extend these subtyping efforts to the clinic and we will argue that preclinical models recapitulate CMS subtypes and can be of vital use to increase our understanding of treatment response and resistance and to discover novel targets for therapy.

Benefit of adjuvant chemotherapy on recurrence free survival per consensus molecular subtype in stage III colon cancer.

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The consensus molecular subtype (CMS) classification divides colon tumors into four subtypes holding promise as a predictive biomarker. However, the effect of adjuvant chemotherapy on recurrence free survival (RFS) per CMS in stage III patients remains inadequately explored. With this intention, we

selected stage III colon cancer (CC) patients from the MATCH cohort ($n = 575$) and RadboudUMC ($n = 276$) diagnosed between 2005 and 2018. Patients treated with and without adjuvant chemotherapy were matched based on tumor location, T- and N-stage ($n = 522$). Tumor material was available for 464 patients, with successful RNA extraction and CMS subtyping achieved in 390 patients (surgery alone group: 192, adjuvant chemotherapy group: 198). In the overall cohort, CMS4 was associated with poorest prognosis (HR 1.55; $p = .03$). Multivariate analysis revealed favorable RFS for the adjuvant chemotherapy group in CMS1, CMS2, and CMS4 tumors (HR 0.19; $p = .01$, HR 0.27; $p < .01$, HR 0.19; $p < .01$, respectively), while no significant difference between treatment groups was observed within CMS3 (HR 0.68; $p = .51$). CMS subtyping in this non-randomized cohort identified patients with poor prognosis and patients who may not benefit significantly from adjuvant chemotherapy.
